What Has Transcranial Magnetic Stimulation Taught Us About Neural Adaptations To Strength Training? A Brief Review

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Abstract

Kidgell, DJ and Pearce, AJ. What has transcranial magnetic stimulation taught us about neural adaptations to strength training? A brief review. *J Strength Cond Res* 25(11): 3208–3217, 2011—The evidence for neural mechanisms underpinning rapid strength increases has been investigated and discussed for over 30 years using indirect methods, such as surface electromyography, with inferences made toward the nervous system. Alternatively, electrical stimulation techniques such as the Hoffman reflex, volitional wave, and maximal wave have provided evidence of central nervous system changes at the spinal level. For 25 years, the technique of transcranial magnetic stimulation (TMS) has allowed for noninvasive supraspinal measurement of the human nervous system in a number of areas such as fatigue, skill acquisition, clinical neurophysiology, and neurology. However, it has only been within the last decade that this technique has been used to assess neural changes after strength training. The aim of this brief review is to provide an overview of TMS, discuss specific strength training studies that have investigated changes, after short-term strength training in healthy populations in upper and lower limbs, and conclude with further research suggestions and the application of this knowledge for the strength and conditioning coach.

Key Words silent period, surface electromyography, motor-evoked potential, corticospinal excitability

Introduction

The well-observed phenomenon of increases in an individual's strength, particularly in the first 4 weeks of starting a strength training program, are generally explained using the neural adaptation paradigm (68). This paradigm proposes that the central nervous system (CNS) is responsible for increases in strength, typically characterized by a noticeable absence of muscular hypertrophy (for reviews, see [2,9,30]). However, little neurophysiological evidence has supported the neural adaptation paradigm. Indirect evidence such as activation patterns between agonist, antagonists, and synergists; or changes in electromyography (EMG) patterns have been used to support neural adaptations (1,26,32,70). Recently, however, studies using the technique of transcranial magnetic stimulation (TMS) have provided the neurophysiological data to support this paradigm. This brief review will present this evidence to inform the strength and conditioning coach that early strength adaptations are predominantly neural in nature. Understanding how the manipulation of acute strength training variables modulates neural adaptations is important when considering the practical implications of this research. Further, TMS data have provided us with the evidence that different groups of muscles respond differently to training load, repetition velocity, and precision of movement. Importantly, understanding the neural responses and adaptations after various training regimes can assist the strength and conditioning coach in developing targeted and effective strength and conditioning programs.

The training-related changes in muscle strength that occur after various strength training regimes arise as a result of improved neural drive to the trained musculature. Increases in specific tension after strength training are thought to occur through improved "efficiency" in descending neural pathways because of adjustments in the activation of agonists and synergists and decreases in coactivation of antagonists (9,19). These adaptations after strength training occur at 2 levels: (a) a supraspinal level, which involves changes in corticospinal excitability and inhibition and (b) spinal level, which involves changes in spinal motoneurons and inhibitory and excitatory interneurons (2,26,30,32,70). Determining the adaptations that occur at a spinal level often involves testing electrically evoked reflexes, such as the Hoffman reflex (H-reflex), which measures the synaptic efficacy of the 1a afferent to the spinal α-motoneuron pool and is modulated by presynaptic inhibition of 1a afferents (60). The volitional wave (V-wave)

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is another electrophysiological technique that measures the magnitude of the efferent drive from the α-motoneuron pool and is influenced by changes in motoneuron firing rate, efficiency of the 1a afferent-motoneuronal synapse, and changes in the intrinsic properties of the motoneuron pool itself (64). Several studies have shown increased H-reflex and V-wave amplitude after strength training, supporting a spinal level of adaptation (1,19,29). Furthermore, the mechanism for enhanced spinal cord reflexes (H-reflex) appears to be attributed to changes in spinal interneurons that integrate the sensory-motor signals and synapse onto the motoneurons controlling the trained muscle (58). For further discussion on these specific mechanisms, the reader is directed to reviews by Semmler and Enoka (71) and Folland and Williams (30).

Until recently, there has been limited evidence to support supraspinal mechanisms underpinning strength gains because of limitations in techniques to access the human primary motor cortex (26). One technique that can provide reliable data of the strength of neural projection between brain and muscle is TMS. Transcranial magnetic stimulation has been successfully used to investigate the adaptations that occur in the CNS after various conditions including skill acquisition (61,62) and degenerative neurological conditions (52,53). Transcranial magnetic stimulation is now being employed to assess changes in the efficacy of neural transmission along the corticospinal pathway after strength training (10,34), yet little is known in the wider strength and conditioning community about TMS and its value in contributing toward strength and conditioning research. We acknowledge that there are many studies that have investigated the question regarding neural adaptations with strength training, which can be found elsewhere (for reviews see [2,12,26]). However, the purpose of this brief review is to provide the strength and conditioning coach with information on the mechanism and technique of single pulse TMS and its role as a research tool in our understanding of neural adaptations after strength training.

Transcranial Magnetic Stimulation

Noninvasive brain stimulation was first developed in 1982 (4,55). Merton and Morton (55) demonstrated that a brief, high voltage electrical stimulus could noninvasively activate axons of corticospinal neurons within the human motor cortex to produce a muscle response. In 1985, Barker et al. (4) showed that it was possible to stimulate the brain using TMS with little or no pain, because TMS bypasses scalp musculature to activate the axons of interneurons that synapse with corticospinal neurons of the primary motor cortex (4,8). Commercially available magnetic stimulators suitable for human study have been available since the mid-1980s (4). There are 2 basic types of magnetic stimulators: single- or paired-pulse TMS, and repetitive TMS (rTMS). The rTMS employs the use of trains of stimulation pulses (up to 100 Hz) and is used primarily in psychology and psychiatry. Single and paired-pulse TMS is used in the research and clinical diagnostic setting, and will be the method discussed in this review. The mechanisms and principles of TMS are well covered in reviews by Hallet (35–37). However, briefly, TMS employs time varying magnetic fields that induce electrical currents in conductive neural tissue, with the induced electric field being proportional to the rate of the magnetic field. If the current induced by the electric field is of a sufficient amplitude and duration, it will depolarize the axons of interneurons that synapse onto corticospinal neurons, recorded and measured as a motor-evoked potential (MEP) in the surface electromyogram (sEMG) of the target muscle.

Single pulse TMS has been shown to have minimal risk to healthy adults (77) and children (24,25,65,67), making TMS very safe (35). Magnetic coils come in many different shapes and sizes. The 2 most common coils are the circular coil, providing relatively powerful fields; or "figure-of-eight" coils (sometimes also called "butterfly coils" [Figure 1]) producing a relatively weaker, yet more focal field at the intersection of the 2 round coils. Figure-of-eight coils have been used extensively for upper limb muscle investigations targeting the hand, forearm, bicep and deltoid muscles (47,61,62,83). Circular coils have also been used to study upper limb muscles, but have also been used for accessing lower limbs (34,66). Large double cone coils are less common than the figure-of-eight and circular coils but have been used to access trunk and leg muscles (73,74,77).

There are several quantifiable components with TMS. The motor threshold (MT) is the minimum amount of stimulation required to produce an MEP. The MT may be observed in both

Figure 1. Participant wearing a fitted cap with markings of 1 cm distance in both antero-posterior (AP) and medio-lateral (M-L) directions. A figure of 8 coil (70 mm) is held tangential to the skull in an antero-posterior orientation to ensure appropriate current flow for activation the motor cortex.

resting muscle or during a tonic voluntary contraction and may be termed active motor threshold or AMT (47). Tonic contraction of the target muscle can reduce the threshold to evoke an MEP, because of increased excitability of neurons at the cortical and spinal level (21–24). The relationship between the stimulus intensity and motor output can be measured by recording stimulus–response curves of MEP amplitude vs. stimulus intensity (of 5% steps of stimulator output), from subthreshold to well above MT (13,20). At stimulus levels well above threshold, the MEP plateaus despite increasing stimulus intensity. This is termed MEP 'saturation' or 'max' (MEP_{max}). The relationship between TMS intensity and MEP amplitude is sigmoidal, and the peak slope of the curve is a measure of the physiological strength of corticospinal projection to the motoneuron pool and reflects the balance between inhibitory and excitatory inputs to the spinal motoneuron pool (13,14).

The measurable components of the MEP are as follows: latency, amplitude, and post-MEP silent period (SP) duration. The latency of the MEP is a reproducible measure of the corticospinal conduction time and measured from the time of stimulation to the onset of the MEP (Figure 2) (35). In healthy individuals, latency is a reliable measure; however, in people with neurological conditions, latency can be variable. Further, it is also well known that tonic voluntary activation (VA) can reduce the latency time of an average of 2–3 milliseconds because of facilitation of spinal motoneurons (69).

The MEP is a relatively synchronous muscle response comprising a series of corticospinal volleys (35). It has been suggested that TMS initially evokes a D-wave (direct excitation of corticospinal neurones) followed by a series of I waves (indirect activation via cortical interneurons) at interpeak latencies of a 1- to 2-millisecond range (35). At the level of spinal motoneurons, D and I waves generate monosynaptic excitatory postsynaptic potentials, which summate to bring α-motoneurons to firing threshold. The result is a contraction (or twitch) in the muscle, which can be recorded as an MEP and is usually quantified by measuring the peak-to-peak amplitude of the MEP "spike" waveform on the sEMG. The corticospinal volley (action potential) underlying the MEP is also influenced by direct (monosynaptic) and indirect (disynaptic and oligosynaptic) inhibitory and excitatory corticospinal connections descending onto the motoneuron pool at the time of stimulation (8,68). Similar to latency, facilitation of a muscle will reduce the threshold for an MEP to occur, lowering the intensity of stimulation applied to the participant. Individuals respond differently to TMS; that is the MEP response for similar stimulus intensity between individuals will differ in amplitude (80,81). However, when controlled for torque and type of motor task, the MEP is a reliable intraparticipant measure (13,43,78), allowing for confident interpretation of changes after acute or chronic interventions. However, it is not uncommon to see investigators using an MEP/M-wave ratio to normalize individual MEP variations between participants (45,65).

In facilitated muscle contractions, the MEP will be followed by a distinctive period of sEMG nonactivity (flat line), termed the SP. The SP can last anywhere between 50 and 300 milliseconds after stimulation and provides a measure of the strength of inhibition in the corticospinal pathway (82,83). It is now recognized that the SP arises from a combination of peripheral (spinal cord) and cortical inhibition and with single pulse TMS the duration of the SP is thought to reflect γ-aminobutyric acid ($GABA_b$)–mediated inhibitory processes (16,82).

Transcranial Magnetic Stimulation Compared to Imaging Techniques and Limitations of Single Pulse Transcranial Magnetic Stimulation

The techniques of functional magnetic resonance imaging (fMRI) and positron emission tomography allow for spatial resolution, providing evidence of areas of cortical activation during real or imagined task performance (35). There are a number of studies that have directly shown increased excitability in a cortical motor area during skill acquisition or, conversely, using TMS to transiently suppress regions of the cerebral cortex, providing stronger evidence of motor regions directly involved in a task (38,41,42,61). TMS has been pivotal in a variety of research areas within neurophysiology and motor control. Many studies now combine both spatial imaging techniques such as fMRI with temporal techniques such as TMS to investigate areas of the brain directly involved with motor function (17,49).

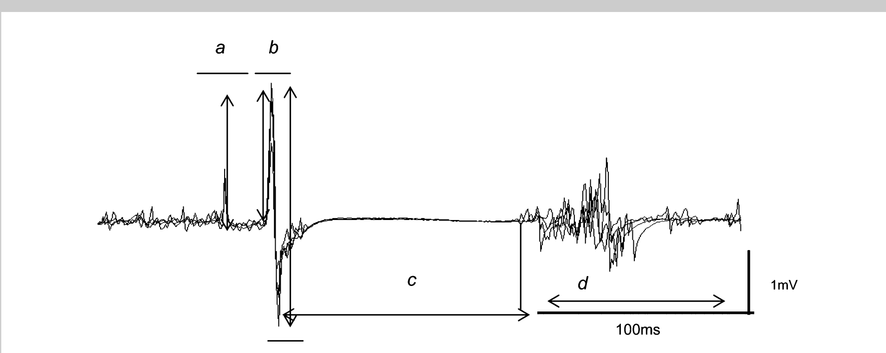

Figure 2. Descending corticospinal volleys recorded from the BB and the resultant MEPs after TMS. Latency duration is measured from stimulus artifact to MEP onset and is shown at (a), peak to peak MEP amplitude is shown at (b). Silent Period duration is measured from onset of MEP to return of EMG at (c). Return of EMG activity is shown at (d).

Although TMS has contributed to considerable advances in CNS physiology, the processes underlying TMS are not simple, and TMS research is not without its limitations (8), particularly with resolution of stimulation focus and protocols used in measuring changes in the corticospinal system after an intervention. However, if these caveats are observed in the research design and measurement protocols, then the findings from TMS research can be interpreted with confidence and reducing the variability in reported findings.

Transcranial Magnetic Stimulation and Strength Training Research

Despite TMS being available since the mid-1980s, it was not until the early 21st century that TMS was used to investigate corticospinal changes after a bout of strength training (14). However, in recent years, there has been a growing interest in the use of TMS to measure neural adaptations after strength training in a range of hand, upper and lower limb muscles. These studies are outlined in Table 1.

Transcranial Magnetic Stimulation Investigations of Hand and Upper Limb Strength Training

Carroll et al. (14) provided the first investigation to examine the effect of short-term strength training on corticospinal excitability. Reporting a 33% increase in muscle strength, these authors found no change in resting MEP amplitude after 4-weeks of dynamic strength training of the first dorsal interosseous muscle. They also observed a reduction in the amplitude of the MEP after training when elicited by both TMS and transcranial electrical stimulation (TES). Because TMS excites the axons of interneurons that synapse onto corticospinal neurons and TES excites corticospinal neurons directly, the authors concluded that strength training altered the functional properties of the spinal cord but not the motor cortex or corticospinal tract. Jensen et al. (42) had participants perform heavy-load (1 repetition maximum [1RM] was not reported) dynamic strength training of the bicep brachii (BB) muscle 3 times per week for 4 weeks. Participants completed 5 sets of 6–10 repetitions (applying progressive overload) over a 4-week training period and training led to a 31% increase in maximal strength. However, a significant reduction in MEP_{max} and calculated slope of the stimulus–response curve measured at rest during stimulation after 4 weeks of strength training was reported.

A recent approach to determine the sites of adaptations with the nervous system is to use TMS to stimulate the primary motor cortex and to also use cervocomedullary stimulation of the brainstem (10) and record the size of the TMS evoked MEP and evoked twitch force. The benefit of this type of stimulation paradigm is that it allows investigators to identify the site or level of adaptation (i.e., cortical or spinal). After 4 weeks strength training, Carroll et al. (10) demonstrated that there was an increase in TMS evoked twitch responses up to 25% of MVC force but not at force levels above 50% of maximal voluntary contraction (MVC). Interestingly, the size of the TMS and cervocomedullary MEP was smaller after the strength training intervention, which was consistent with the findings from the earlier study performed by Carroll et al. (14). The findings from these articles support improved synaptic efficiency at the level of the spinal cord (10,14). Further, these data also support a training-related increase in neural drive to the trained musculature at low force levels. However, at high force levels, Carroll et al. (10) reported no change in MEP amplitude and TMS-evoked twitch response. Similar findings have also been reported by Lee et al. (50) with an increase in MVC force by 11.0 ± 8.7%, however, with no change in TMS-evoked twitch force. It is not clear why strength training did not alter MEP and twitch amplitudes at moderate to higher forces (above 25% of MVC). Nonetheless, changes at a cortical level cannot be ruled out.

Studies to date have used TMS to identify the neural adaptations to strength training and have only reported data on the MEP amplitude. Neural activation of muscle is dependent on a combination of both the excitatory and inhibitory processes (45). Recently, Kidgell and Pearce (45) not only presented MEP amplitude but also SP duration data. These authors had participants complete 12 supervised strength training sessions over a 4-week training period that involved 6 sets of 10 repetitions of 3-second maximal isometric contractions against a force transducer, interspersed with a 3-second rest, with 3-minute rest between sets. The MEPs and maximal M-waves (delivered with supramaximal electrical stimulation of the ulnar nerve) were applied before and after the strength training intervention at 10% above AMT during a constant isometric contraction of 5 and 20% of MVC. Mean strength in the training group increased by 34%, and when normalized as a percentage of the maximum M-wave, there were no significant differences in the size of the MEP between time and groups at 5 and 20% MVC. However, the novel finding was that the observation of a significantly reduced SP duration in the strength training group of 16 milliseconds at 5% MVC and 25 milliseconds at 20% MVC. Further, correlation analysis showed a significant moderate inverse correlation ($r = -0.51$) between the change in strength and the change in mean SP duration for the strength training group. The data suggest that short-term maximal isometric strength training (4 weeks), improves strength, because of decreased neural inhibitory mechanisms. The practical applications of the findings of this study show that the net corticospinal drive to the motor neuron pool was improved as a result of a reduction in cortical inhibition (45).

Kidgell et al. (47) have also recently presented MEP amplitude and SP duration after short-term heavy-load, controlled-speed strength training of the BB. Unlike previous studies in the BB muscle (42), these authors observed a 28% increase in strength after 4 weeks of heavy-load strength training that resulted in a 53% increase in MEP amplitude, without associated increase in muscle mass. However, no change in SP duration was observed. The differences in

Table 1. Summary of specific single pulse TMS and strength training intervention research outlined in this review.

Reference	Participants	Muscle(s)	Strength training intervention	TMS change	Strength change (%)
Carroll et al. (14)	1 Female sets and 15 male sets (8 controls, 8 trained) age range: 22–36 y	First dorsal interosseous	Isotonic: 6 reps × 4 sets at 70% MVC 3/wk × 4 wks, training load increased by 5% when 6 reps × 4 sets could be completed	Reduced corticospinal excitability at 40–60% of MVC torque	33
Jensen et al. (42)	11 Females and 13 males (8 controls, 8 skill trained, 8 strength trained) age (mean ± *SD*): 25 ± 5 y	Biceps brachii	Isotonic: 10–6 reps × 5 sets 3/wk × 4 wks	Reduced corticospinal excitability (41.5% decrease)	31
Griffin and Cafarelli (34)	1 Females and 19 males (10 controls, 10 trained) age range: 18–32 y	Tibialis anterior	Isometric: 10 reps × 6 sets × 5-s MVC 3/wk × 4 wks	Increased corticospinal excitability (32%)	18
Beck et al. (5)	17 Males, 10 females (10 controls, 9 balance trained, 8 strength trained) age range: 20–38 y	Tibialis anterior and soleus	Isometric: 4 × 10 dorsiflexion and plantarflexion at 30–40% of MVC 4/wk × 4 wks, training load increased every 2 wks	Increased cortical excitability in ballistic trained (9.5%); reduced cortical excitability in stability trained (4%)	9
Lee et al. (50)	7 Females and 16 males (11 controls, 12 trained) age range: 18–51 y	Wrist abductors	Isotonic: 4 × 8 at 70% of MVC 3/wk × 4 wks, training load increased by 5% every 2 wks	No change corticospinal excitability	11
Kidgell and Pearce (45)	3 Females and 13 males (8 controls, 8 trained) age (mean ± *SD*): 24.12 ± 5.21 y	First dorsal interosseous	Isometric: 10 MVC × 3–6 sets 3×/wk × 4 wks	No change in corticospinal excitability, decrease in corticospinal inhibition of 16 ms at 5% MVC and 25 ms at 20% MVC	34
Lee et al. (51)	7 Females and 13 males (10 controls, 10 trained) age range: 18–24 y	Wrist extensors	Isometric: 10 MVC (1–2 s) × 3×/wk × 4 wks	Increased TMS twitch activation (2.9%)	31.5 In trained limb, 9.3 in untrained limb
Kidgell and Pearce (46,47)	14 Females and 12 males (13 controls, 13 trained) age (mean ± *SD*): 26.8 ± 7.31 y	Biceps Brachii	Isotonic: 4 sets × 6–8 at 80% 1RM 3×/wk × 4 wks	Increased corticospinal excitability (53% strength trained arm, 33% increase in contralateral arm)	28.0 In trained limb, 19.2 in untrained limb

*TMS = transcranial magnetic stimulation; 1RM = 1 repetition maximum; MVC, maximal voluntary contraction.

findings, as suggested by Kidgell et al. (47), may arise from the strength training methodology reported. Carroll et al. (14) progressively overloaded participants from 70 to 85% 1-RM, whereas the study by Kidgell et al. (47) started at 80% of 1RM. Although each repetition was reported by Carroll et al. (14) to be performed slowly, the exact repetition timing was not provided, even though repetition timing has been shown to be an important component to exercise prescription and strength development (48). Although this study did not directly compare the effect of repetition velocity (i.e., fast vs. slow) on strength development and corticospinal excitability, the data demonstrate that repetition timing may be important for inducing neural adaptations that underpin strength development. Further, by manipulating repetition timing coupled with a heavy training load (80% 1RM) improves neural excitability from the primary motor cortex and along the corticospinal pathway to the motoneuron pool innervating the trained muscles. The net benefit of this is improved force production because of improved motor unit recruitment and activation. Therefore, the strength and conditioning coach should consider manipulating these acute strength training variables to increase strength and modulate adaptations in the CNS.

Transcranial Magnetic Stimulation and Upper Limb Cross-education Strength Studies

Cross-education refers to the strength improvement of muscles in one limb as a result of chronic physical activity of the opposite homologous limb (27,28). Cross-education of muscle strength has been demonstrated comprehensively in many upper limb studies using static, dynamic, and imagined muscle contractions (7,9,39,51,57,84). To date, limited TMS-specific studies have measured cross-education in upper limbs and no lower limb cross-education TMS studies have been published. A meta-analysis by Munn et al. (57) demonstrated on average increases of 25.1% in the trained arm resulted in a mean increase of 7.8% in muscle strength of the contralateral homologous muscle. However, the mechanism for this interlimb interaction remains unclear. It has been suggested that some form of adaptive plasticity within the organization of the contralateral elements of the CNS, such as the corticospinal pathway may be the locus of adaptation (11,27).

Recently, a number of studies have used TMS to measure cortical VA (50,51,73). Understanding VA is important clinically when assessing people with impairments in corticospinal drive and during exercise to identify the sources of fatigue (3,10,30). To generate maximum force, the CNS must activate all motor neurons that innervate the muscle producing the desired force. A failure to activate all the contributing motor units innervating the active muscle will result in a less than maximal effort, and as such VA is considered to be incomplete (3,31,73). Cortical VA is conventionally determined via the twitch interpolation technique and involves the application of a single supramaximal electrical stimulus applied to the motor nerve during the performance of an MVC (3). If additional force is evoked as a result of the electrical stimulus, cortical VA is considered to be incomplete. An alternative method, using TMS has been used to identify the site of failure within the CNS and involves assessing the "extra" force that is evoked by a maximal TMS pulse applied over the primary motor cortex during an MVC (3,6,30,31). Using this technique, Lee et al. (50) demonstrated an increase in cortical VA of the opposite untrained limb after 4 weeks of isometric strength training of the right wrist extensor muscles. Twitch interpolation, using TMS, was used to assess the changes in cortical VA of the untrained wrist extensors. After training, there was a significant increase in strength of the trained (31.5%) and untrained wrist extensors (8.2%), which was accompanied by a significant decrease in twitch amplitude (35%) contributing to a significant increase in VA (2.9%). The finding of reduced amplitude of the superimposed twitch after the strength training period was interpreted as increased corticospinal output to the untrained wrist extensors (49). Kidgell and Pearce (46) demonstrated a 19.2% increase in strength of the untrained BB of the left limb after 4 weeks of heavy-load (80% of 1RM) slow speed (4-second extension and 3-second flexion), dynamic BB strength training of the right limb. Stimulus–response curves were constructed pre and poststrength training and showed for the trained group, in the right motor cortex projecting to the spinal motoneuron pool controlling the untrained left BB, a 33% increase in the amplitude of the MEP at 20% above MT, and a 26.5% increase in maximal MEP amplitude. Collectively, these findings show that the contralateral transfer of strength is mediated by changes in neural transmission along the corticospinal pathway to the untrained limb.

Neural mechanisms underpinning cross-education may reside in ipsilateral projections of the corticospinal pathway (40,65) or via bilateral cortical activation (15,75,76). It has been reported that the size of the ipsilateral corticospinal projection is in the order of 10–30%; however, there is some variation between individuals (18,33,59). Strens et al. (75) used both single pulse TMS and rTMS in an attempt to quantify the role of the ipsilateral motor cortex in force regulation. After bilateral rTMS to both motor cortices, participants were unable to match a target force; however, when the contralateral motor cortex was stimulated repetitively, participants were able to match the target force, demonstrating that the ipsilateral motor cortex and corticospinal pathway plays an important role in the modulation of force. Hortobágyi et al. (40) and Perez and Cohen (65) have demonstrated increased MEP amplitudes in ipsilateral corticospinal projections after wrist flexion of increasing force further supporting the contribution of ipsilateral corticospinal pathways. Therefore, in accordance with Kidgell and Pearce's (46) previous research, the practical application of these data supports the notion of prescribing heavy-load controlled repetition velocity strength training to induce neural adaptations to both the trained and untrained limbs. As such, manipulating the acute strength training variables in conventional strength training practice and during rehabilitation using the practice of cross-education appears to be a viable practical way of increasing strength that is modulated by the corticospinal pathway.

It is well documented that hemispheric asymmetries within the motor cortex are present because of limb dominance (33,54,72). Evidence suggests that the left hemisphere projecting and controlling the dominant right hand (in right-handed individuals) plays an important role in controlling the left limb, more so than the right hemisphere controlling the right limb (27,28,79). This may also explain that the current findings that transfer of strength, in right limb dominant individuals, is stronger from the right limb to the left than vice versa (see Farthing [27] review). However, further research, particularly in testing left limb dominant individuals, is required to test this theory.

A second physiological explanation for increases in corticospinal excitability to the untrained limb after unilateral training may be found in the motor irradiation theory (15,73). Motor irradiation is the proposed mechanism whereby cortical activity during strong or high-intensity unilateral contractions diffuses from the active motor cortex to the "inactive" motor cortex giving rise to bilateral activation of both motor cortices (15). It has been proposed that this occurs through interhemispheric pathways via corticocortical cells within the cerebral cortex in layers 2 and 3 of the primary motor cortex that form a broad, intrinsic horizontal projection system via the corpus callosum (54,56). The observations by Kidgell and Pearce (46) and Lee et al. (51) appear to support some aspect of motor irradiation theory, given the increase in efficiency of corticospinal transmission to the untrained limb. Combined fMRI and TMS studies have also provided physiological evidence showing unilateral motor activity causing bilateral activation of both the left and right motor cortex via increased excitability of existing intrinsic horizontal pathways within the motor cortex (47,63,66).

Similarly, to induce cross-education effects after unilateral training, the strength coach should prescribe heavy-load, slow tempo strength exercises. However, if prescribing for an injured participant, caution should be noted as these studies were conducted in noninjured participants to investigate the neural outcomes of cross-education strength training. Further research is required to investigate cross-education in clinical populations (such as unilateral injured athletes) using exercise paradigms suitable for injured athletes.

Transcranial Magnetic Stimulation Investigations of the Lower Limb

In comparison to hand and upper limb and cross-education training studies, TMS studies investigating neural adaptations in the lower limb after a training intervention have received relatively less attention. However, the studies to date have demonstrated changes in corticospinal excitability after short-term strength training interventions.

Beck et al. (5) compared corticospinal measures between 2 groups performing either postural stability maintenance training or a ballistic ankle strength training task (16 sessions in total) over a 4-week period. Background sEMG levels, M-wave and H-reflex amplitudes remained unchanged in both groups. However, in the ballistic strength training group, there was a 9.5% increase in MEP amplitude in the tibialis anterior (TA) muscle, whereas the postural stability group demonstrated a reduction in MEPs in TA. The authors concluded that MEP activation in the ballistic training reflected increased efficiency of corticospinal transmission to the trained musculature, whereas reduced MEP amplitude in the stability training group reflected improved motor control through cortical inhibitory effects. Griffin and Cafarelli (34) used TMS to investigate corticospinal responses after 4 weeks of strength training of the TA muscle. Ten individuals performed 6 sets of 10 5-second MVC contractions, 3 times per week for 4 weeks (12 sessions in total) and were compared to a nontrained control group. The strength trained group demonstrated a 10 and 18.1% increase in strength at 2 weeks and at the end of the 4-week training period, respectively. The TMS measurements showed significant increases in peak-to-peak MEP amplitudes at 2 weeks (31.6%) and remained significantly higher to baseline, and the control group, at 4 weeks (16.3%). Once again, these data demonstrate enhanced efficiency of neural transmission along the corticospinal pathway after short-term strength training. The practical implications for increased efficiency in corticospinal transmission to the trained musculature may enhance the force output of the muscle by facilitating changes in agonist–synergist–antagonist coordination and enhanced motor unit activation. Once again, these data demonstrate that manipulating the way in which strength training exercises are performed (ballistic and controlled at high force levels) modulates strength gains because of increased corticospinal excitability. This manipulation of strength training variables appears to be consistent for proximal upper limb and lower limb muscles. As such, it is important that the strength and conditioning coach focuses on prescribing exercises that challenge the nervous system. The available TMS data to date suggest that manipulation of the acute strength training variables (resistive load, repetition timing, and precision of movement) is important for modulating adaptations with the CNS.

Practical Applications

The training-dependent increase in strength observed after short-term strength training appears to be modulated by changes in neural efficiency along the corticospinal pathway. Changes in corticospinal control after strength are influenced by the type of strength training program prescribed (isometric and isotonic) and by the manipulation of the acute training variables, especially resistive load and repetition timing. Therefore, the strength and conditioning coach should focus on prescribing exercises that uses a heavy resistive load, with velocity and precision of movement. Although recent studies have tried to elucidate the site of adaptation within the nervous system (i.e., spinal or supraspinal), further research is required. There still remains some conflicting evidence as to the neural adaptations induced which may be explained via the limitations of some single pulse TMS research as outlined by Burke and Pierrot-Deseilligny (8), in particular, the muscle investigated (intrinsic hand vs. gross arm), or the type of training employed (moderate vs. heavy-load intensities). Furthermore, the neural adaptations that occur after strength training most likely involve adaptations at a cortical, spinal, and motor unit level. Nevertheless, this brief review suggests that strength training improves the efficiency of neural transmission along the corticospinal pathway to the trained and untrained (i.e., cross-education) musculature as a consequence of manipulating the acute strength training variables (load, velocity, and precision of movement). Future investigations should look to examine all aspects of the TMS evoked response (latency, MEP amplitude, and SP duration) on comparing adaptations after strength training in fine and gross muscles, the intensity (% of 1RM) of training, velocity of movement, and other forms of resistance training (such as stability and whole body vibration training). Finally, to move this research forward, further studies must employ single pulse TMS with cervicomedullary stimulation to account for the site of adaptation and to reduce the limitations associated with single pulse TMS.

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